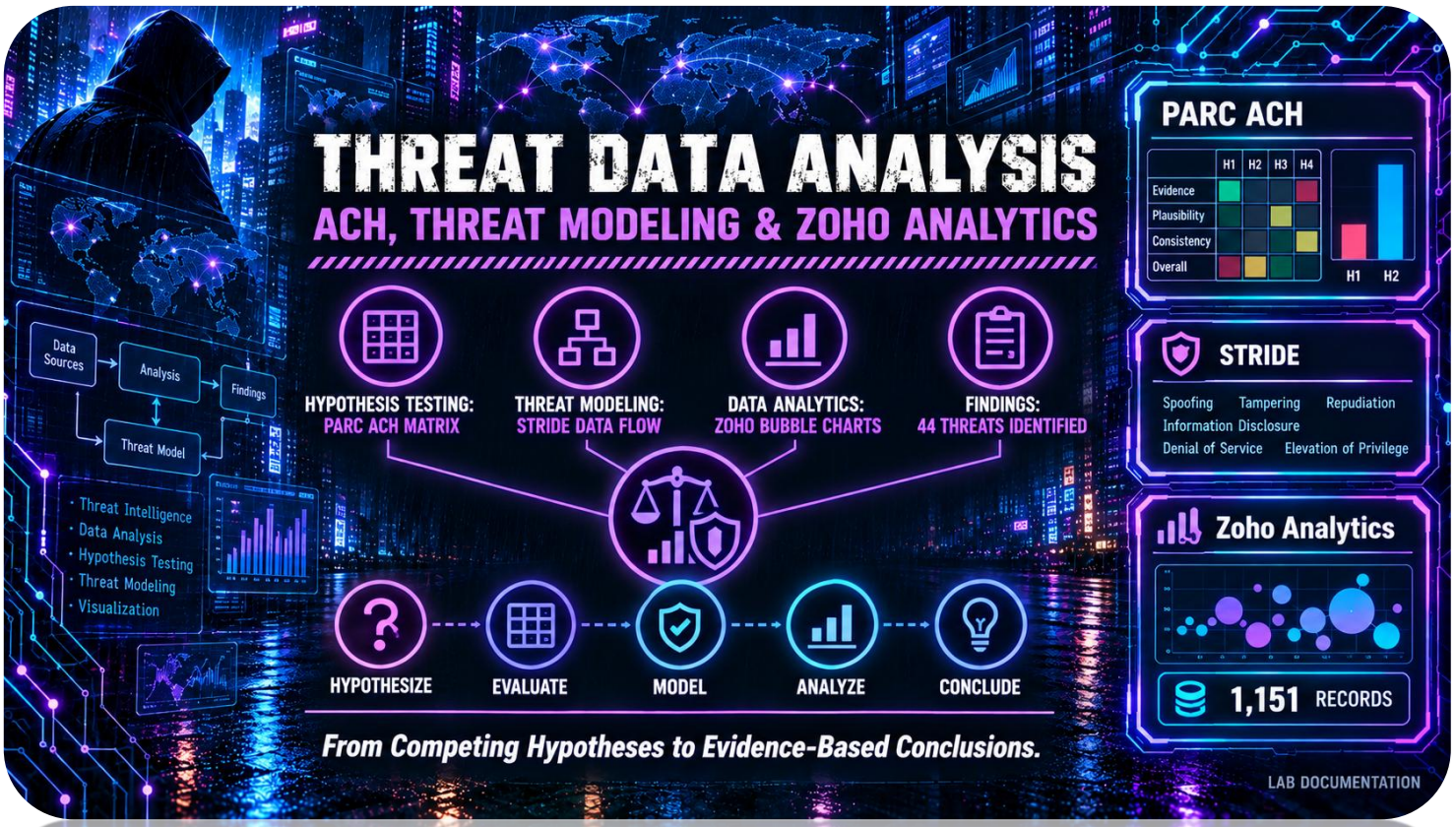


Abdul Wahid

Lab Report

Threat Data Analysis – ACH, Threat Modeling & Zoho Analytics

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Lab 1: Analysis of Competing Hypotheses (ACH) using PARC ACH

Analysis of Competing Hypotheses (ACH) is a structured analytic technique used in intelligence analysis to evaluate multiple hypotheses simultaneously. In this lab, the PARC ACH 2.0.5 tool was used to systematically assess hypotheses related to a cybersecurity incident — specifically whether the EHR vendor was compromised via a malicious patch or whether the database permissions were simply too broad.

The ACH process follows: Hypotheses → Evidence → Diagnostics → Refinement → Inconsistency → Sensitivity → Conclusion and Evaluation.

Step 1: Launching PARC ACH and Reading the First-Time Users Guide

Upon launching PARC ACH 2.0.5, the application presents its main matrix interface. A "First Time Users" dialog box appears, which explains the core methodology: ACH breaks a complex problem into component parts. The analyst fills in a complete set of hypotheses, enters evidence, and rates how each piece of evidence supports or refutes each hypothesis. This tool does not identify the best hypothesis automatically — it guides the analyst through a structured thought process to reach that conclusion independently.

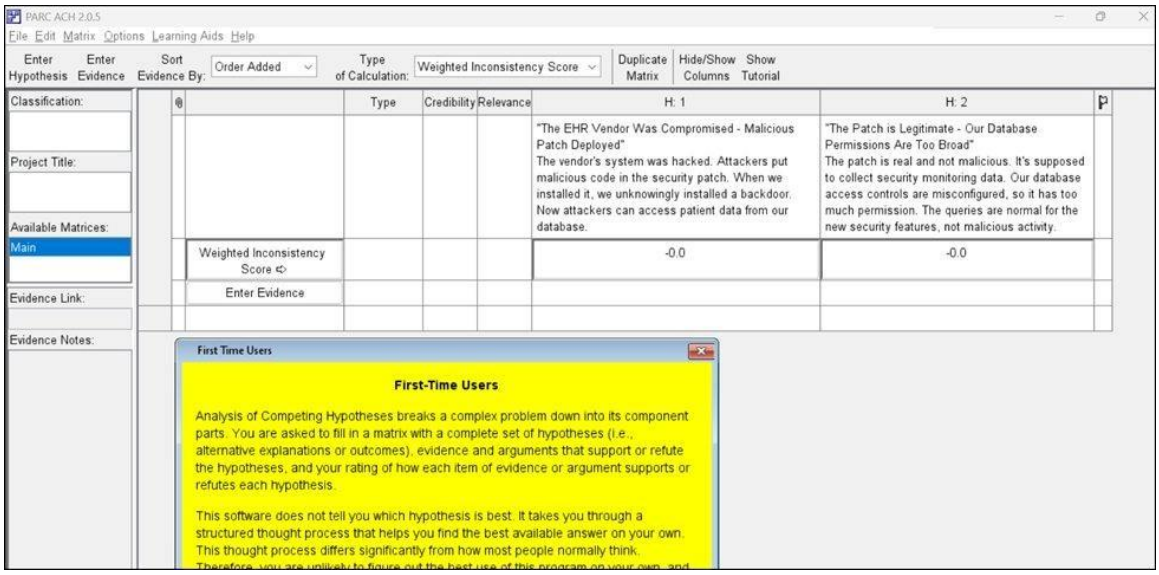


Figure 1.1 – PARC ACH 2.0.5 main window showing the First-Time Users dialog explaining the ACH methodology

Step 2: Entering Hypotheses

Two competing hypotheses were entered into the ACH matrix:

H:1 – "The EHR Vendor Was Compromised – Malicious Patch Deployed": The vendor's system was hacked. Attackers put malicious code in the security patch. When it was installed, a backdoor was unknowingly created, now allowing attackers to access patient data.

H:2 – "The Patch is Legitimate – Our Database Permissions Are Too Broad": The patch is real and not malicious. It collects security monitoring data, but database access controls are misconfigured, giving it too much permission. The queries are normal for the new security features, not malicious activity.

At this stage, both hypotheses show an initial Weighted Inconsistency Score of -0.0, indicating that no evidence has yet been entered or rated.

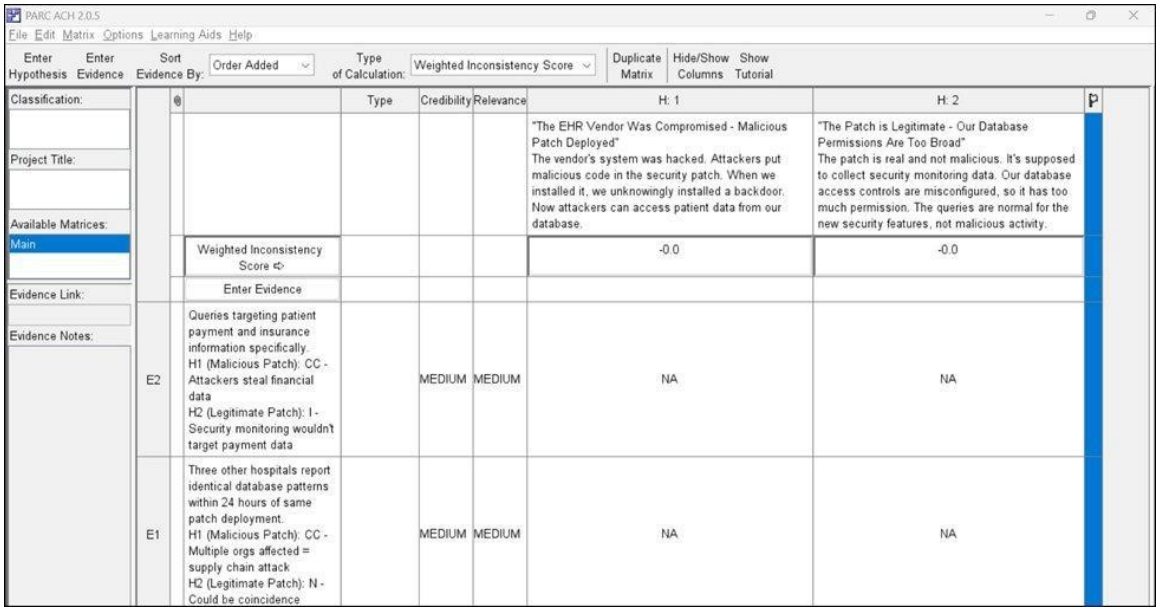


Figure 1.2 – PARC ACH matrix with both hypotheses entered and initial evidence items (E1, E2) added with NA ratings

Step 3: Entering Evidence and Rating Consistency

Evidence items were entered and rated against each hypothesis using the following rating scale:

II = Evidence highly inconsistent with Hypothesis | I = Inconsistent | NA = Not Applicable | N = Neutral | C = Consistent | CC = Evidence highly consistent with Hypothesis

Evidence E2: "Queries targeting patient payment and insurance information specifically." This was rated MEDIUM credibility and HIGH relevance. This evidence is more consistent with the malicious patch hypothesis (H:1), since legitimate security monitoring would not specifically target financial/insurance data. For H:2 (legitimate patch), this is rated as Inconsistent (I) since permission-based monitoring would not selectively query payment records.

Evidence E1: "Three other hospitals report identical database patterns within 24 hours of same patch deployment." This was rated MEDIUM credibility and LOW relevance. For H:1 (malicious patch), this is rated CC (highly consistent) since a supply chain attack would naturally affect multiple organizations simultaneously. For H:2 (legitimate patch), this is rated I (inconsistent) since a legitimate patch would not produce suspicious patterns across multiple hospitals.

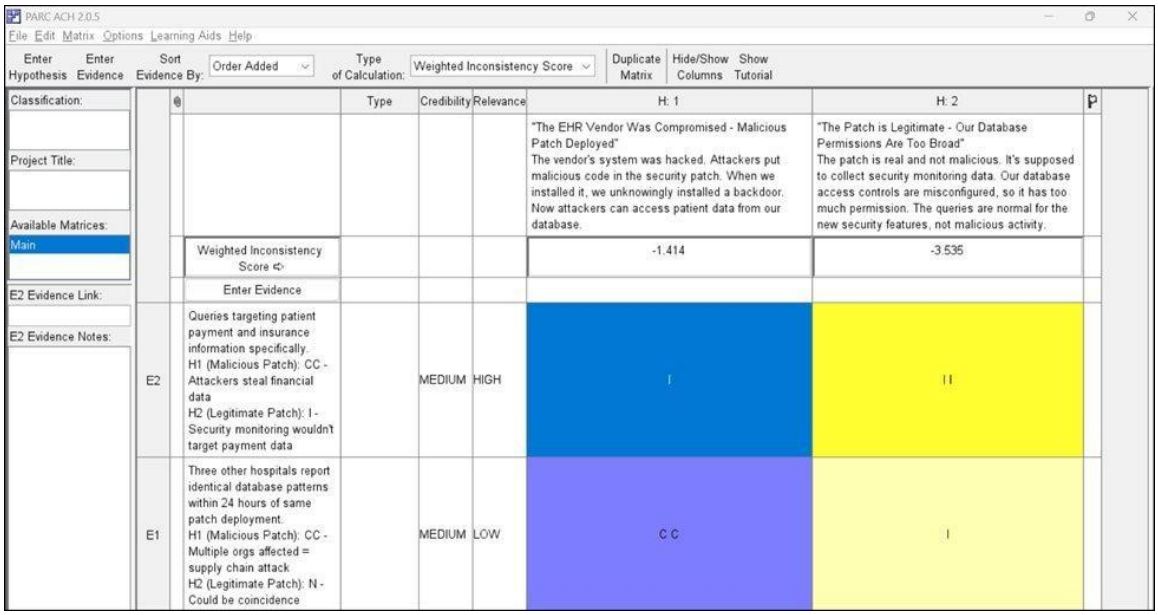


Figure 1.3 – PARC ACH matrix showing evidence ratings applied: H:1 scores -1.414 and H:2 scores -3.535, indicating H:2 has more inconsistencies

Step 4: Accessing the Matrix Menu

The Matrix menu provides key functions: Enter Hypothesis, Enter Evidence, Delete Selected Hypothesis/Evidence, Duplicate Matrix, Hide/Show Columns, and Show Chart. The "Show Chart" option is especially important as it visualizes the weighted inconsistency scores for each hypothesis as a bar chart, making comparison intuitive.

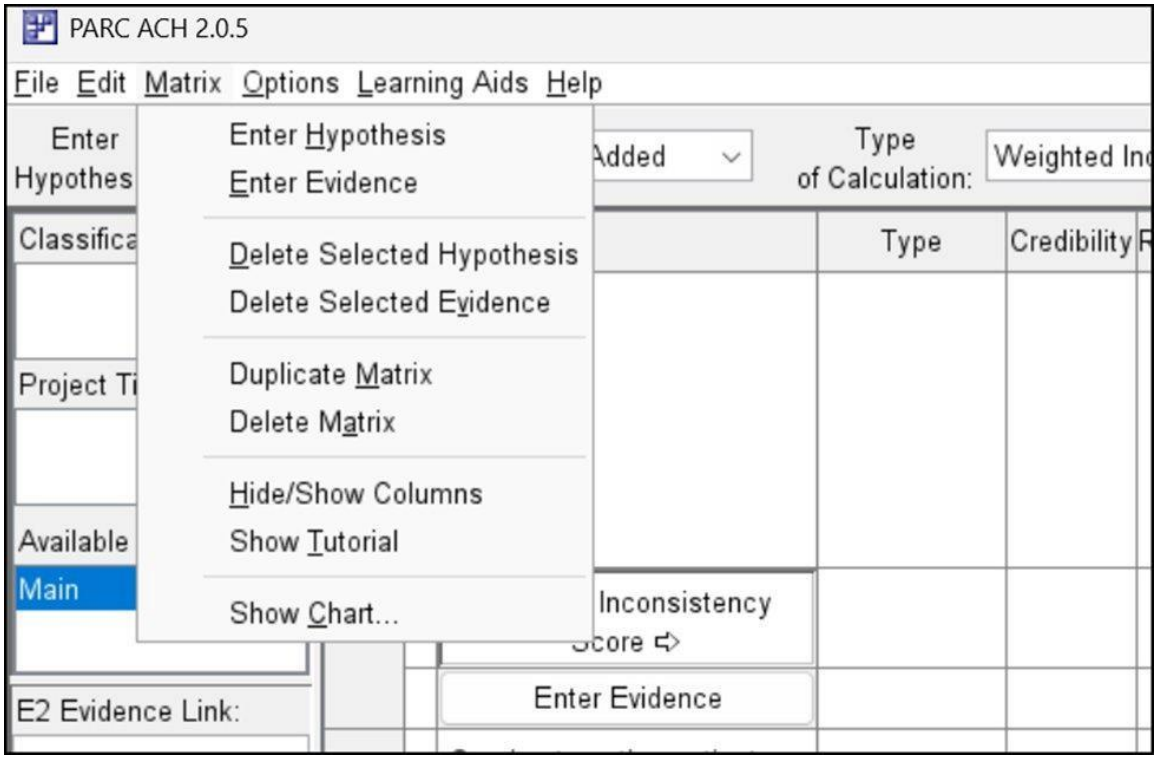


Figure 1.4 – PARC ACH Matrix menu showing available options including Show Chart for visual analysis

Step 5: Visualizing Results – Inconsistencies per Hypothesis

After all evidence was entered and rated, the Matrix → Show Chart option was used to generate the Inconsistencies per Hypothesis bar chart. The chart clearly shows:

H:1 (Malicious Patch – red bar): Score ~1.41 — significantly lower inconsistency, meaning this hypothesis is more consistent with the available evidence.

H:2 (Legitimate Patch/Broad Permissions – blue bar): Score ~3.54 — much higher inconsistency, indicating this hypothesis contradicts more of the evidence.

In ACH, the hypothesis with the LOWEST weighted inconsistency score is considered most likely correct. Therefore, H:1 — that the EHR vendor was compromised and a malicious patch was deployed — is the most supported conclusion based on the collected evidence.

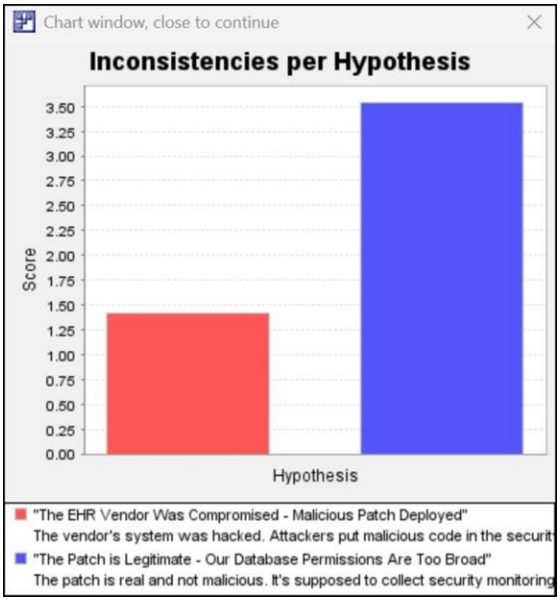


Figure 1.5 – Inconsistencies per Hypothesis chart: H:1 (red, ~1.41) has fewer inconsistencies than H:2 (blue, ~3.54), making H:1 the more probable hypothesis

Lab 2: Threat Modeling using Microsoft Threat Modeling Tool 2016

Threat modeling is a structured methodology for identifying, analyzing, and ranking potential security threats within a system. Microsoft Threat Modeling Tool 2016 uses the STRIDE framework (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege) to systematically identify threats in data flow diagrams. This lab demonstrates building a threat model from scratch and analyzing a prebuilt WebRequestModel.

Step 1: Microsoft Threat Modeling Tool 2016 – Home Screen

The tool opens with a structured home screen providing three main options: Create A Model (model a system by drawing diagrams), Open A Model (analyze threats in an existing model), and a Getting Started Guide. The Threat Modeling Workflow describes four steps: Select template → Create data flow diagram → Analyze model for potential threats → Determine mitigations.

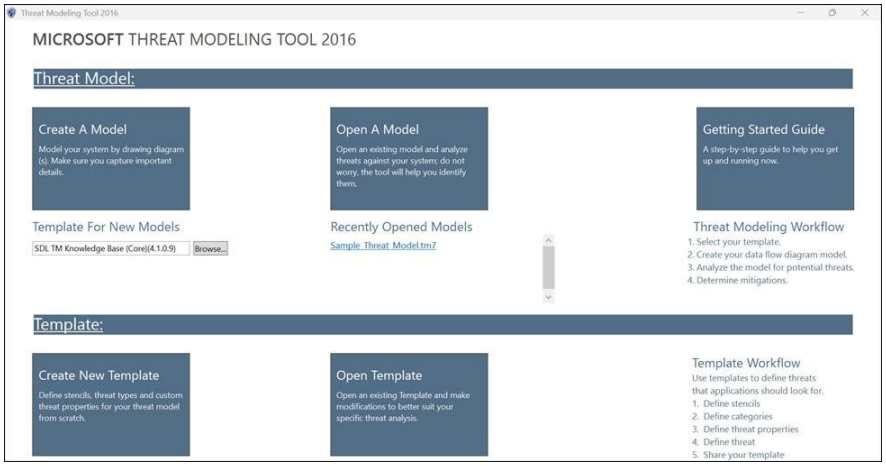


Figure 2.1 – Microsoft Threat Modeling Tool 2016 home screen showing options to Create or Open a model

Step 2: Adding a Human User Stencil to the Diagram

A new model was created by clicking "Create A Model." The tool opens a blank diagram canvas with a Stencils panel on the right. To model an external user interacting with a web application, the "Human User" stencil was selected from the Generic External Interactor category and dragged onto the drawing sheet. The Element Properties panel shows this element's name and attributes (Type: Human). The Messages panel at the bottom confirms no issues yet.

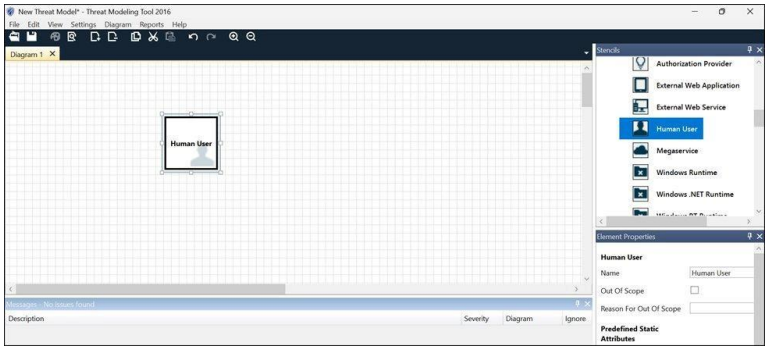


Figure 2.2 – Human User stencil added to the diagram canvas; Element Properties shows its type as Human

Step 3: Adding an External Web Application Stencil

An "External Web Application" stencil was also added from the Generic External Interactor category. This represents the web-facing component that will interact with the Human User. Both entities are now placed on the diagram canvas, ready to be connected via data flow connectors. The stencil library shows several other available interactor types: Browser, Authorization Provider, External Web Service, Megaservice.

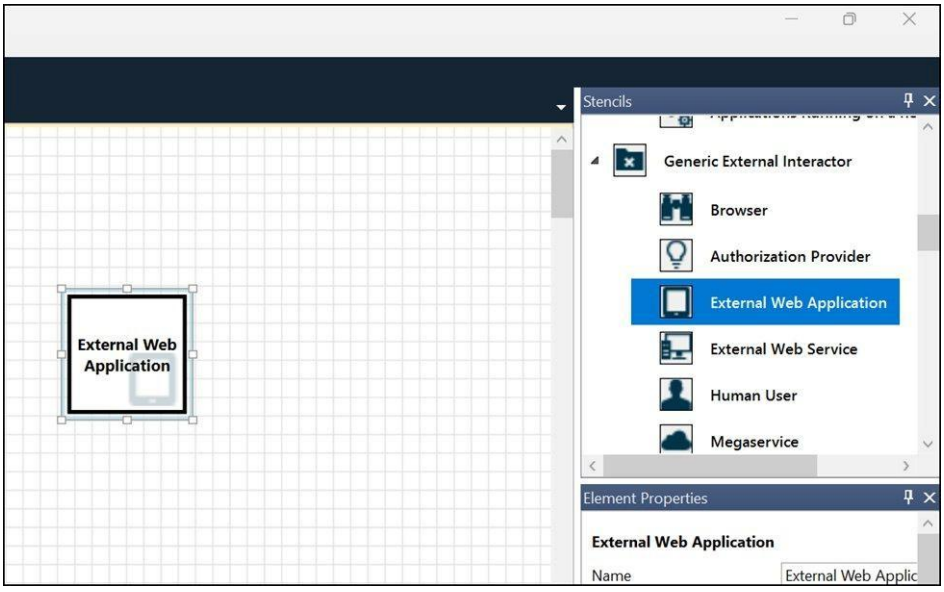


Figure 2.3 – External Web Application stencil added to the diagram alongside the Human User element

Step 4: Adding an HTTP Data Flow Connector

From the Generic Data Flow stencil category, an "HTTP" connector was dragged onto the canvas. This represents unencrypted HTTP traffic between entities. The connector appears as an arrow object labeled "HTTP" on the canvas. The data flow stencil options visible include HTTP, HTTPS (encrypted), Binary, and IPsec, each representing a different communication protocol with different security properties.

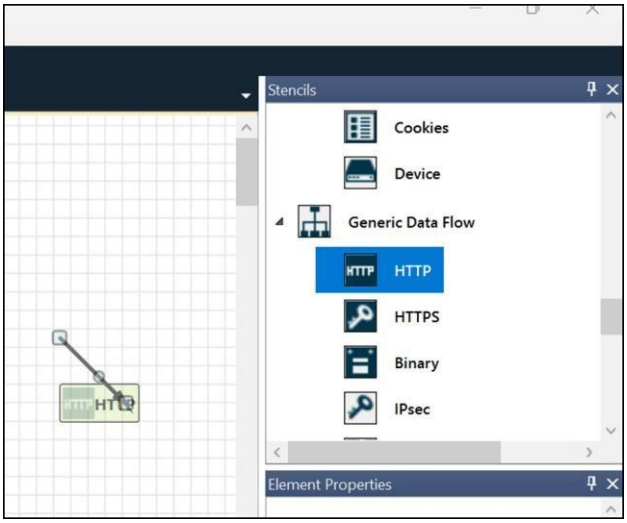


Figure 2.4 – HTTP data flow connector dragged from the Generic Data Flow stencil panel onto the canvas

Step 5: Connecting Human User to External Web Application (One Direction)

The HTTP connector was connected so that its tail connects to the Human User and its head points to the External Web Application. This models a one-directional request from the user to the application. The arc-shaped arrow clearly shows the directionality of the data flow. At this stage, only one direction of communication is modeled.

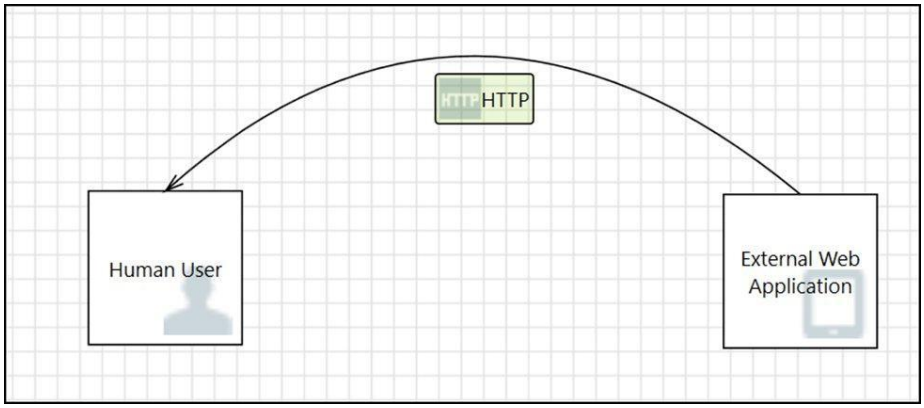


Figure 2.5 – HTTP connector showing one-way data flow from Human User to External Web Application

Step 6: Adding the Return HTTP Data Flow

A second HTTP connector was added in the reverse direction — from the External Web Application back to the Human User. This models the response flow. Both arcs together create a bidirectional communication model, with one arc for the request and one for the response, forming a complete request-response cycle between the user and the web application.

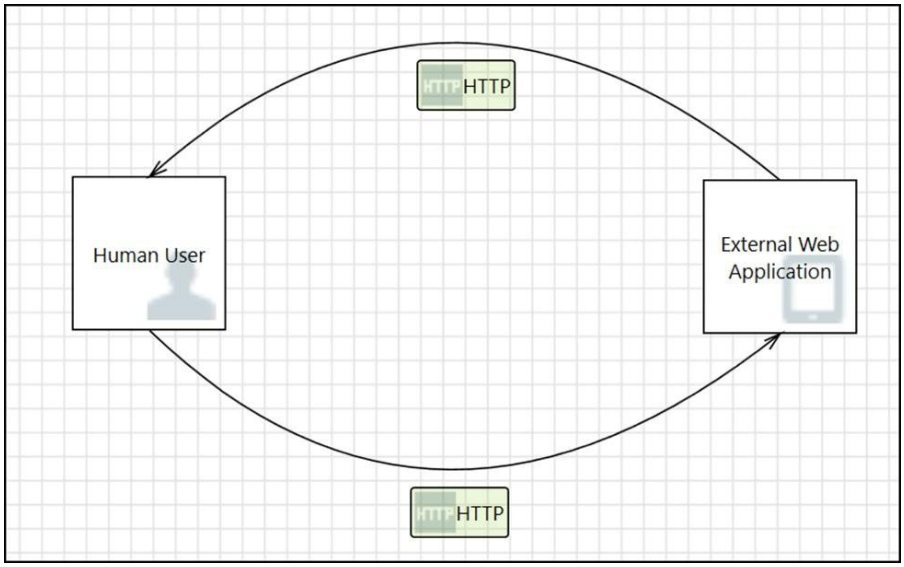


Figure 2.6 – Two HTTP connectors now model bidirectional communication: request (top arc) and response (bottom arc)

Step 7: Renaming the HTTP Request Connector

To improve clarity and specificity, the first HTTP connector was renamed to "HTTP Request" using the Element Properties panel on the right side. The name appears on the connector label in the diagram. This naming convention helps distinguish the request and response flows when the threat tool generates automatic threat analysis, as threats are associated with specific named interactions.

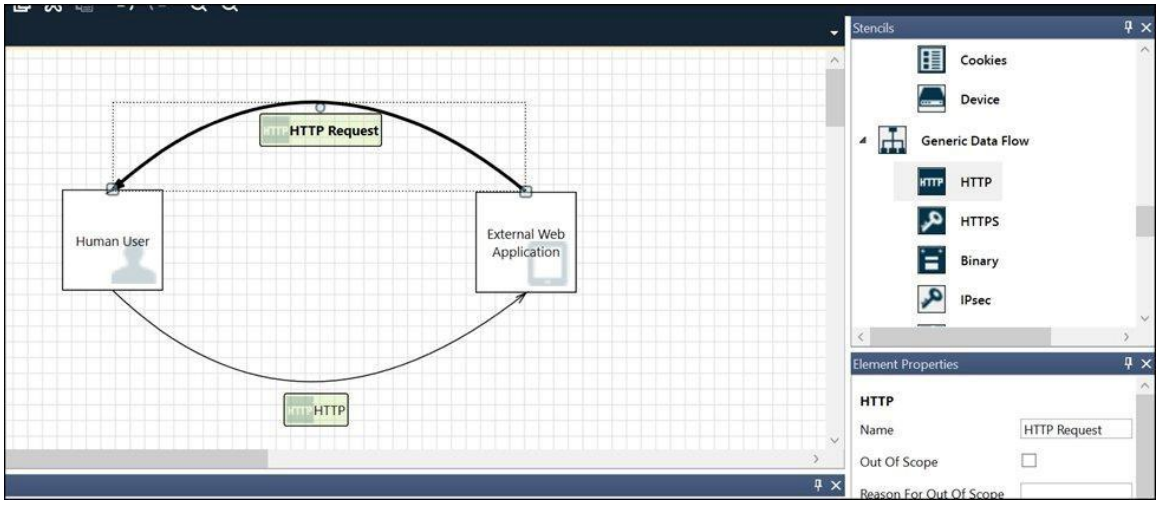


Figure 2.7 – Top HTTP connector renamed to "HTTP Request" via the Element Properties Name field

Step 8: Completed Basic Model – HTTP Request and Response

The complete basic model shows Human User on the left connected to External Web Application on the right via two labeled connectors: "HTTP Request" (top arc, user to application) and "HTTP Response" (bottom arc, application to user). This models a simple web request lifecycle. From this diagram, the tool will automatically identify STRIDE threats applicable to each element and data flow.

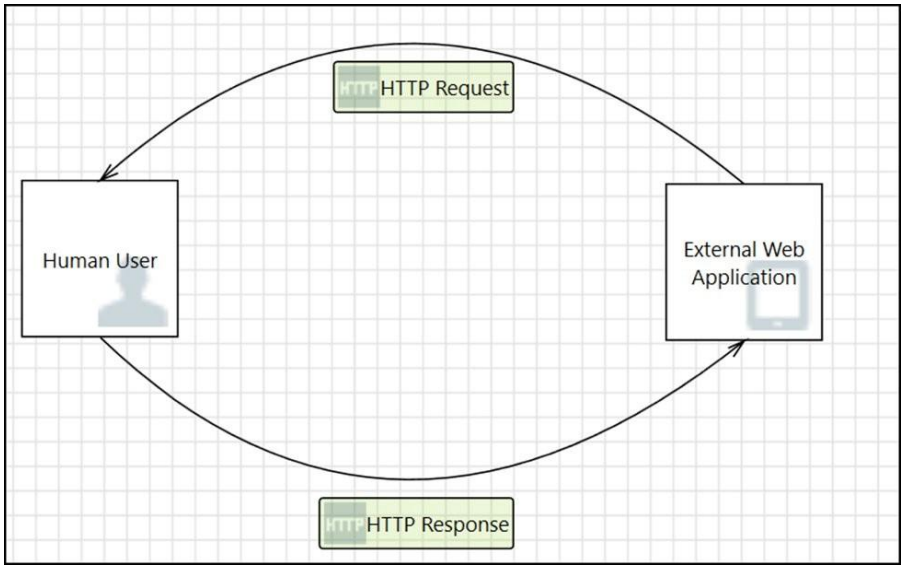


Figure 2.8 – Completed basic threat model showing HTTP Request and HTTP Response flows between Human User and External Web Application

Step Y: Opening a Saved Model (File → Open)

To demonstrate threat analysis on a more complex model, the File → Open menu was accessed. This allows loading any saved .tm7 threat model file. The menu also shows other options like Save, Save As, Apply Template, Threat Model Information, and Close Model. The keyboard shortcut Ctrl+O provides quick access.

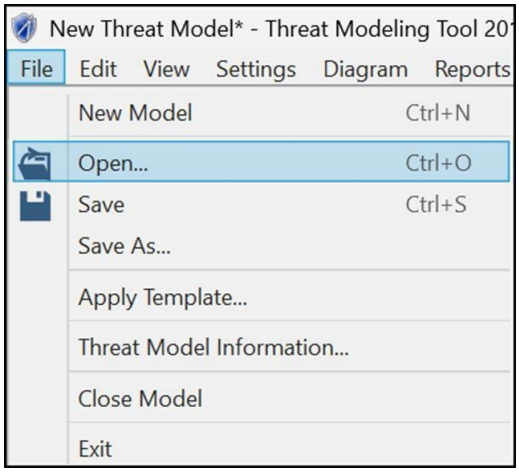


Figure 2.9 – File menu showing the Open option (Ctrl+O) to load a prebuilt threat model

Step 10: The WebRequestModel File

The file "WebRequestModel.tm7" (a TMT7 data file, 7/5/2018) was selected from the file system for opening. The .tm7 format is the native file format for Microsoft Threat Modeling Tool 2016. This prebuilt model contains a more complex system architecture representing a realistic web request flow across multiple components.



Figure 2.10 – WebRequestModel.tm7 file selected; this is a prebuilt complex threat model from July 2018

Step 11: WebRequestModel – Complex Data Flow Diagram

The WebRequestModel diagram represents a complete web architecture: User → Web Application → Web Server → Cloud Storage. The data flows are labeled with their protocols: HTTP Request/Response between User and Web Application; UDP Query Request/Response between Web Application and Web Server; UDP Data Request/Response between Web Server and Cloud Storage. This multicomponent model enables comprehensive threat identification across the entire request pipeline.

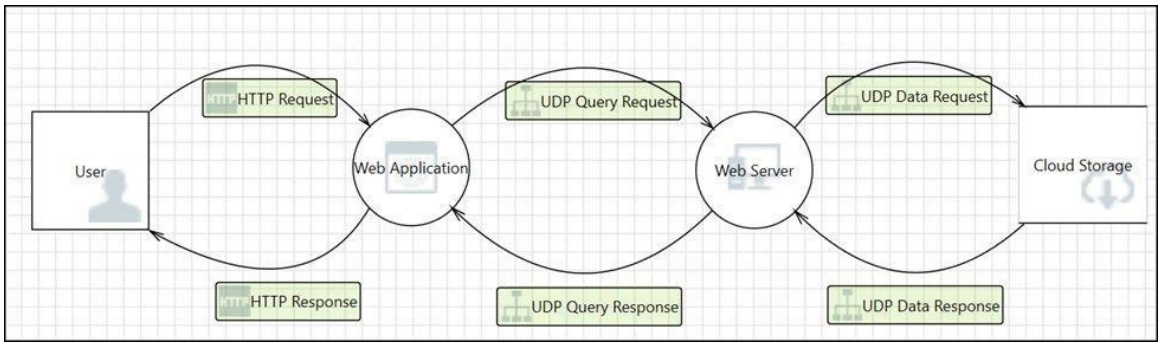


Figure 2.11 – WebRequestModel complex architecture showing User, Web Application, Web Server, and Cloud Storage with labeled data flows

Step 12: Threat Model Information

Via File → Threat Model Information, the model's metadata was filled in with project details. This metadata is included in all generated reports for traceability and ownership documentation.

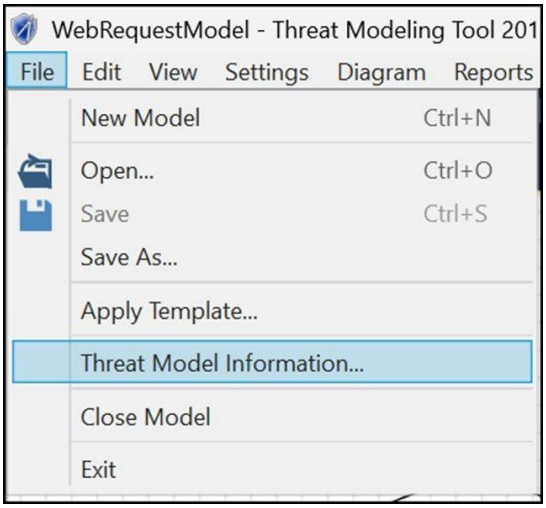


Figure 2.12 – File menu showing "Threat Model Information..." option for entering model metadata

The Threat Model Information dialog captures:

Threat Model Name: Web Request Model | Owner: Abdul Wahid | Contributors: Analyst & Team | Reviewer: Management & Clients | High Level System Description: Sample Threat Model | Assumptions: No Current Assumptions | External Dependencies: Not Applicable | Template: SDL TM Knowledge Base (Core) v4.1.0.9

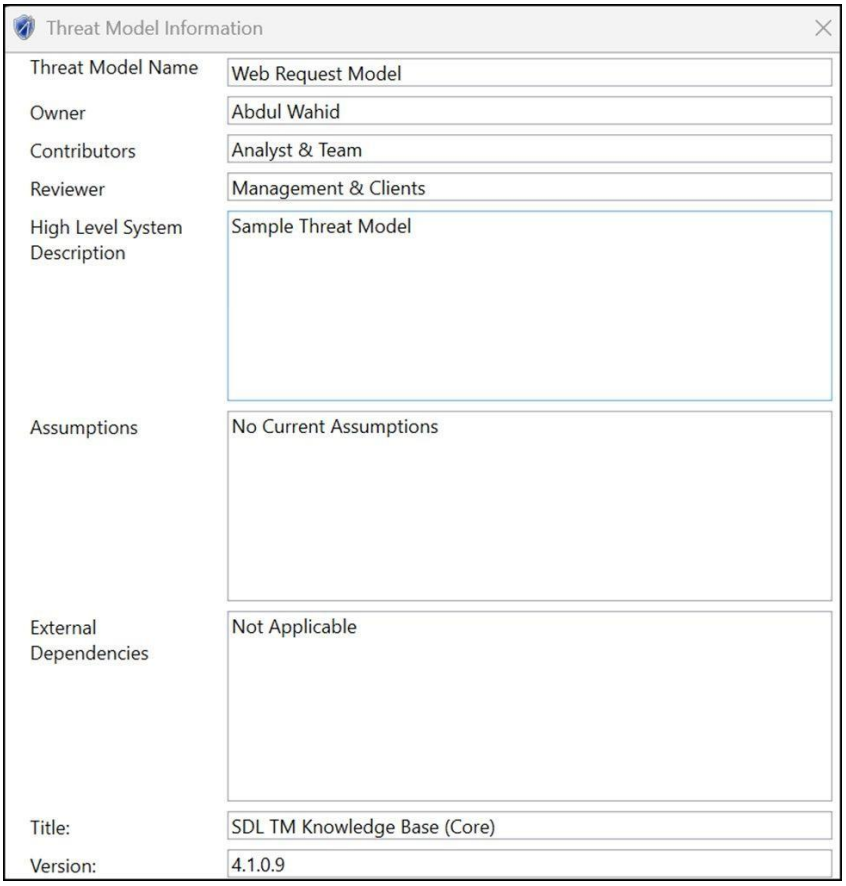


Figure 2.13 – Threat Model Information dialog populated with project metadata including owner and reviewer details

Step 13: Generating the Full Report

To perform threat determination and identification, the report generation feature was used via Reports → Create Full Report. This generates a comprehensive HTML threat analysis report documenting all identified threats, their categories, states, and priorities across all elements and interactions in the diagram.

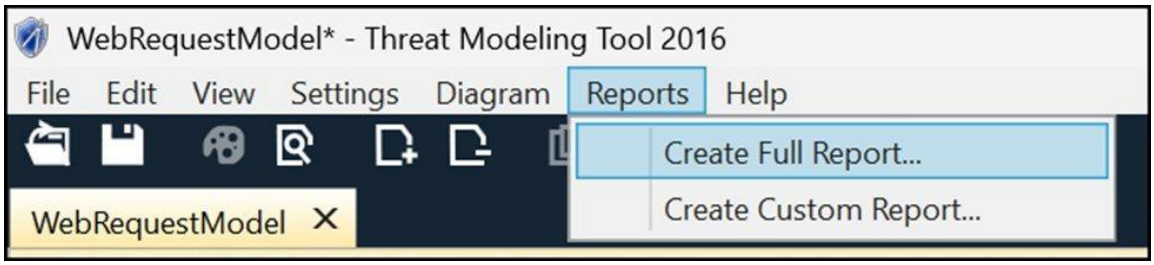


Figure 2.14 – Reports menu showing "Create Full Report" option for generating the complete threat analysis document

In the Generate Report dialog (Custom Threat Properties), the "Generate Report" button was clicked to produce the HTML output. This step completes threat determination — the tool automatically applies the STRIDE model to each data flow and element, generating threat entries based on the diagram components.

Figure 2.15 – Generate Report dialog with "Generate Report" button to produce the full HTML threat model report

Lab 2 (Continued): WebRequestModel Threat Modeling Report Results

Threat Modeling Report – Summary Page

The generated Threat Modeling Report (saved as CTI report.htm) opened in the web browser and displays the complete threat analysis. The report header shows: Created: 5/3/2026 5:04:42 PM | Threat Model Name: Web Request Model | Owner: Abdul Wahid | Reviewer: Management & Clients | Contributors: Analyst & Team | Description: Sample Threat Model | Assumptions: No Current Assumptions | External Dependencies: Not Applicable.

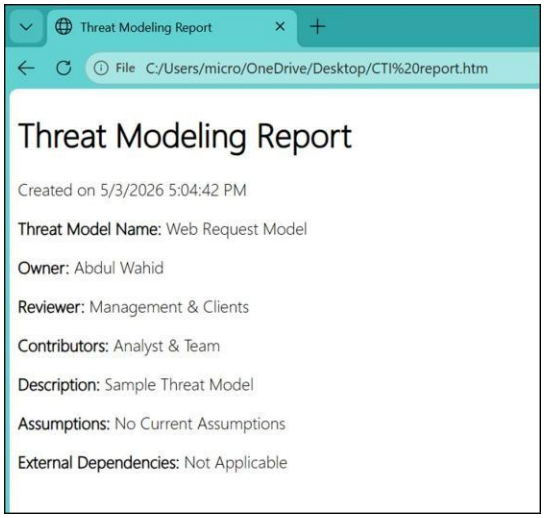


Figure 2.16 – Threat Modeling Report header showing model metadata as seen in the browser

Threat Model Summary Statistics

The report's summary section provides a quantitative overview of all identified threats in the WebRequestModel:

Not Started: 38 threats | Not Applicable: 0 | Needs Investigation: 4 | Mitigation Implemented: 2 | Total: 44 threats | Total Migrated: 0

This means 44 distinct threats were automatically identified by the STRIDE model. Of these, 4 require immediate investigation and 2 already have mitigations in place. The remaining 38 are identified but not yet addressed, representing the security backlog for this system.

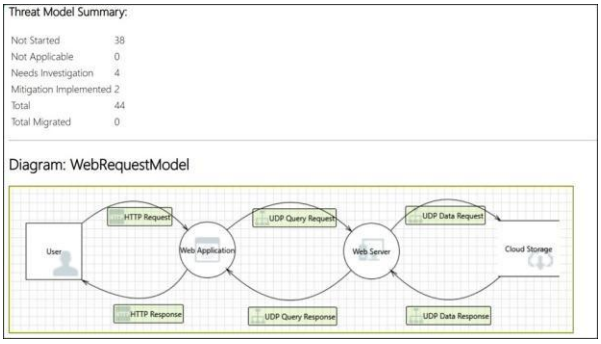


Figure 2.17 – Threat Model Summary showing 44 total threats: 38 Not Started, 4 Needs Investigation, 2 Mitigations Implemented

WebRequestModel Diagram Summary

The diagram-level summary mirrors the overall model statistics since this model contains only one diagram (WebRequestModel). The same 44 threats are distributed across all the interactions modeled in the diagram: HTTP request/response, UDP query flows, and cloud storage data flows. Each interaction type generates different categories of threats based on the STRIDE methodology.

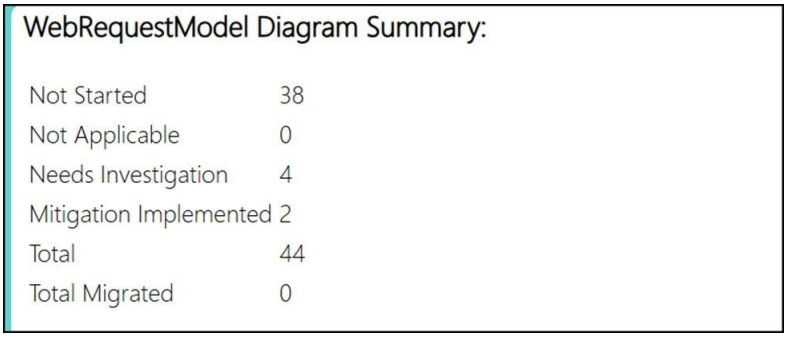


Figure 2.18 – WebRequestModel Diagram Summary confirming 44 threats including 4 Needs Investigation and 2 Mitigations Implemented

Threats Not Associated with an Interaction (Items 1–4)

The first four threat entries in the report are nonstandard threats not associated with a specific interaction. Threats 1, 2, and 4 are User-defined category threats with "Needs Investigation" state and High priority — these were custom threats added to address conditions specific to this deployment. Threat 3 is a Denial of Service category threat, also Needs Investigation/High priority. All four currently have no mitigation provided and no description filled in, indicating they require analyst attention.

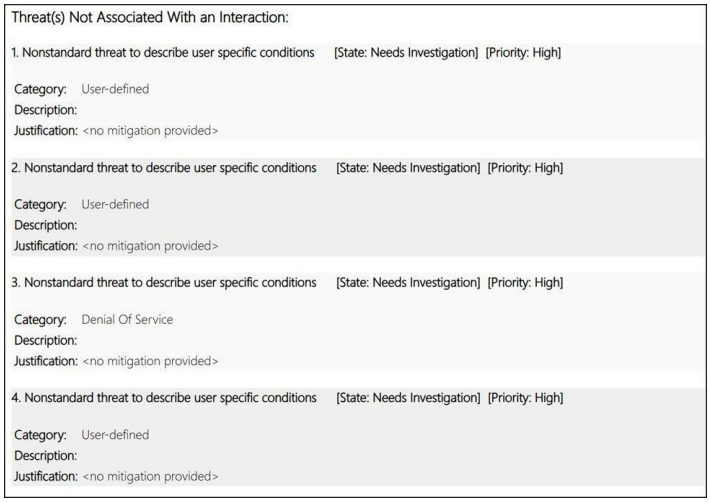


Figure 2.19 – Threats 1–4: Nonstandard user-specific threats including Denial of Service, all requiring investigation with no mitigations yet

Threats 5–8: Not Started, High Priority

Threats 5 through 8 are also nonstandard user-defined threats, but with "Not Started" state, indicating they have been identified but investigation has not yet begun. All are High priority and have no descriptions or mitigations provided. This pattern is typical in initial threat models where the STRIDE process has flagged potential issues but the security team has not yet had time to investigate or document mitigations.

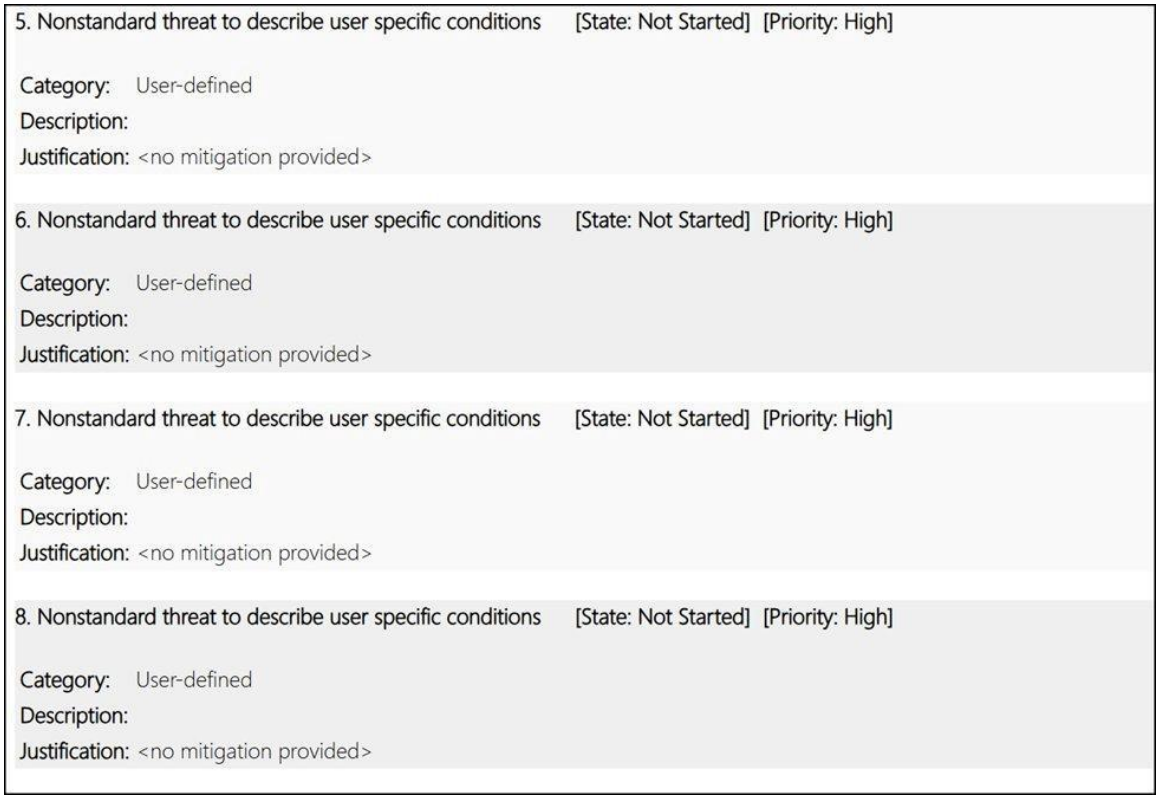


Figure 2.20 – Threats 5–8: Additional Not Started/High Priority user-defined threats awaiting investigation

Lab 3: Data Analysis using Zoho Analytics (Zoho Reports)

Data analysis is a critical component of threat intelligence — transforming raw data into actionable insights. In this lab, Zoho Analytics (formerly Zoho Reports) was used to import and analyze the HHS Cybersecurity Breaches dataset (HHSCyberSecurityBreaches.csv), a real-world dataset containing 1,151 records of healthcare data breach incidents reported to the U.S. Department of Health and Human Services. The analysis demonstrates how an analyst can use visual analytics to extract threat intelligence from bulk data.

Step 1: Signing into Zoho Analytics

The analyst signed into Zoho.com using student credentials (243699@students.au.edu.pk). The Zoho home page displays available apps including Mail, CRM, Books, and Desk. From here, the analyst navigated to the Analytics application (formerly Zoho Reports) to begin the data analysis process.

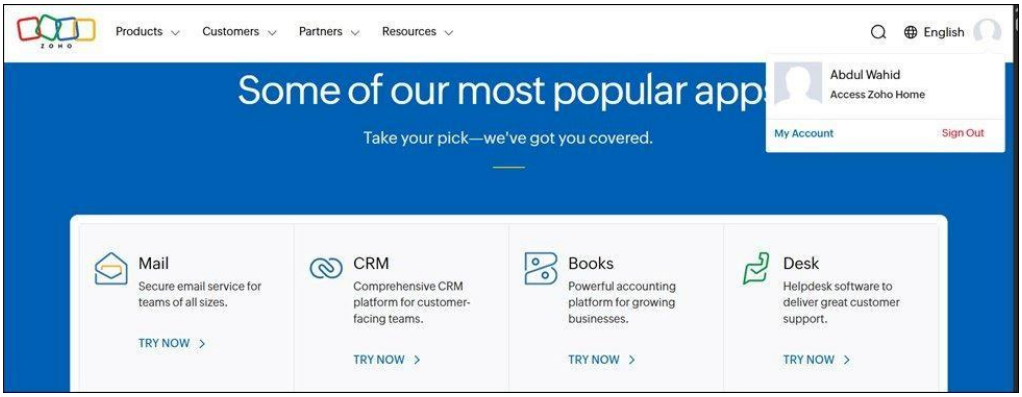


Figure 3.1 – Zoho home page showing the analyst signed in as Abdul Wahid (243699@students.au.edu.pk)

Step 2: Zoho Analytics Loading Screen

Zoho Analytics loaded using the student account credentials. The loading screen shows the application logo and the student email address, confirming successful authentication. Zoho Analytics is a cloud-based business intelligence platform that supports data import from multiple sources, chart creation, pivot tables, and dashboard building.

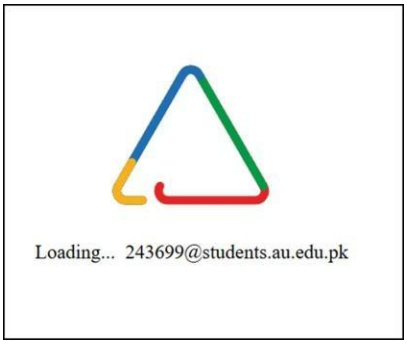


Figure 3.2 – Zoho Analytics loading screen confirming login with student credentials (243699@students.au.edu.pk)

Step 3: Zoho Analytics Dashboard – No Dashboards Yet

After logging in, the Zoho Analytics interface shows a clean dashboard view. Since no data has been imported yet, the message "You do not have any dashboards" appears. The interface shows the Premium Trial status (14 days remaining) and an "Import Your Data" button to begin the data analysis workflow. The left navigation includes Explorer, Dashboards, Reports, Data, and More sections.

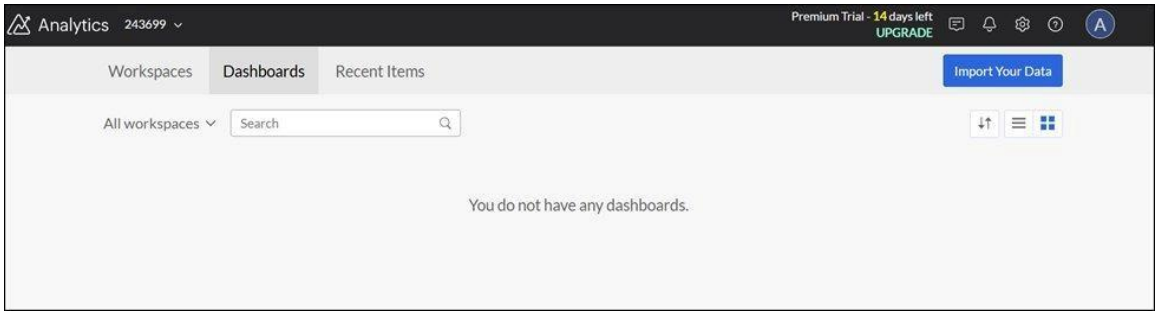


Figure 3.3 – Zoho Analytics Dashboards view showing no dashboards yet; ready to import data

Step 4: Import Your Data – Source Selection

Clicking "Import Your Data" reveals the wide range of data source options available in Zoho Analytics. The analyst selected "Files" to import the local CSV dataset. Other available sources include Feeds/URLs, Cloud Storage/Drive, Databases & Datalakes, Elasticsearch, Streaming Data, Zoho CRM, Zoho SalesIQ, Salesforce, Bigin, Microsoft Dynamics 365, and Google Analytics 4 — demonstrating the tool's versatility for threat intelligence data aggregation.

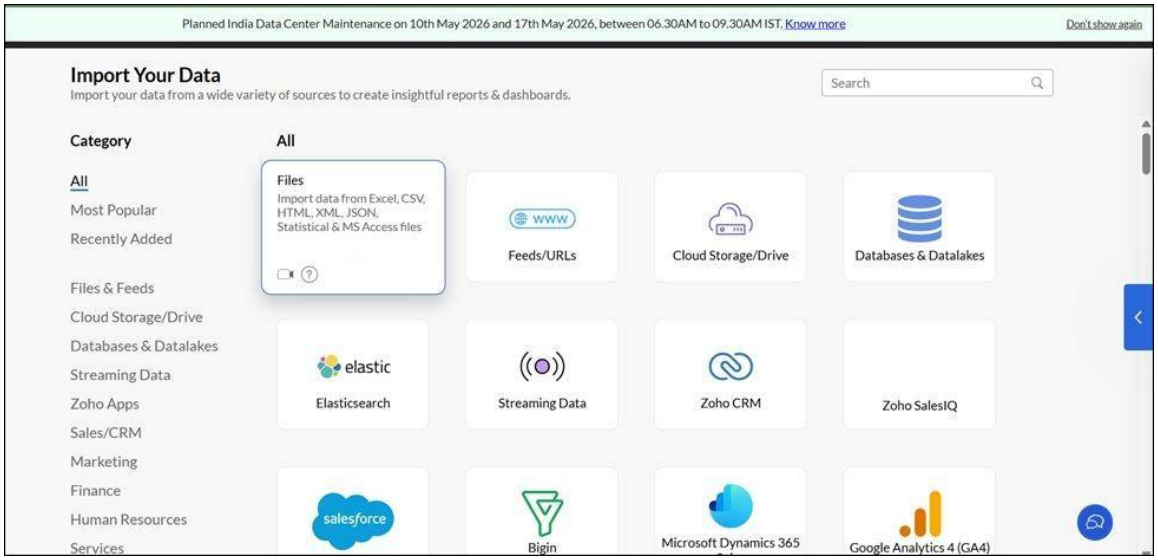


Figure 3.4 – Import Your Data page showing available sources; "Files" was selected for CSV import

Step 5: Configuring the File Import (Step 1 of 2)

In the Step 1 of 2 import dialog, the following configuration was set:

Workspace Name: BULK-Data-File | Description: "Add the File for Analysis" | File Type: CSV, TSV & Other Text Format | Data Location: Local Drive | File: HHSCyberSecurityBreaches.csv

The CSV file was selected from the local drive. General Info on the right confirms data size constraints. The workspace name "BULK-Data-File" organizes this dataset within the analytics environment.

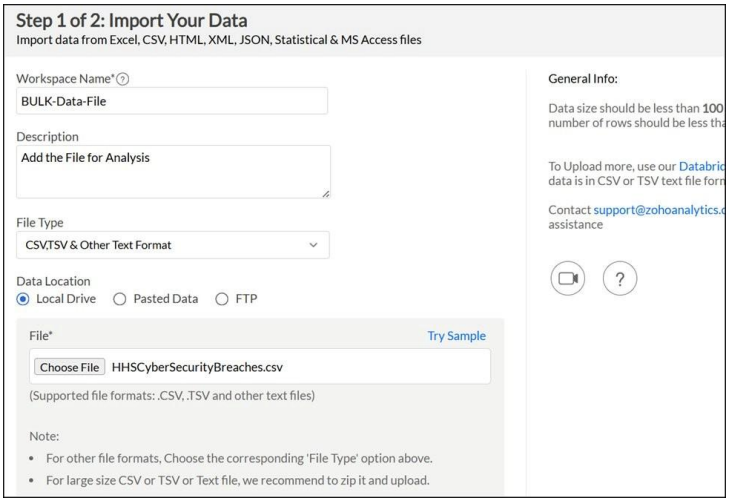


Figure 3.5 – Step 1 of 2: Import configuration with workspace name, description, and CSV file selected

Step 6: Data Preview and Column Configuration (Step 2 of 2)

The Step 2 preview shows the HHSCyberSecurityBreaches table configuration: Table Name: HHSCyberSecurityBreaches | First Row Contains Column Names: Yes | Date Format: yyyy-MM-dd. The data preview shows 5 sample rows with columns: Name of Covered Entity (Plain Text), State (State/Province), Covered Entity Type (Plain Text), Individuals Affected (Positive Number), Breach Submission Date, Type of Breach, Location of Breached Information, Business Associate Present, and Web Description. The dataset spans healthcare providers, health plans, and business associates across all U.S. states.

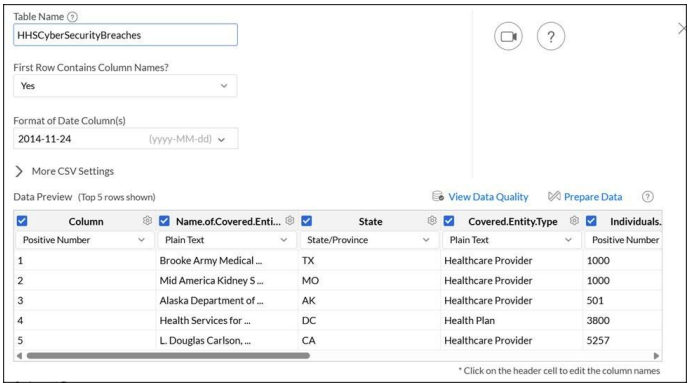


Figure 3.6 – Step 2 data preview showing column types and sample records; 10 columns detected automatically

Step 7: The HHSCyberSecurityBreaches CSV File Reference

The source file "HHSCyberSecurityBreaches.csv" (Excel CSV format, 319 KB, created 7/4/2018 3:32 PM) contains the complete dataset of healthcare data breaches reported to the U.S. Department of Health and Human Services Office for Civil Rights. This is real-world threat data used by security analysts to identify patterns, trends, and high-risk entities in healthcare cybersecurity.

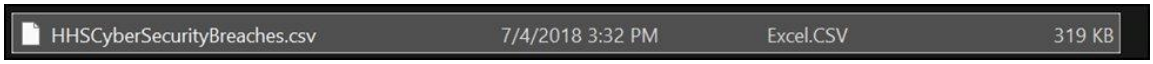


Figure 3.7 – HHSCyberSecurityBreaches.csv file shown; 319 KB Excel CSV file containing the breach dataset

Step 8: Import Error Handling and Create

The import settings include an "On Import Errors" option set to "Set Empty Value for the Column" — this ensures that any data formatting errors (such as malformed dates or invalid numbers) do not abort the entire import. Instead, problematic cells are set to empty/null. After confirming these settings, the "Create" button was clicked to initiate the data import and table creation in Zoho Analytics.

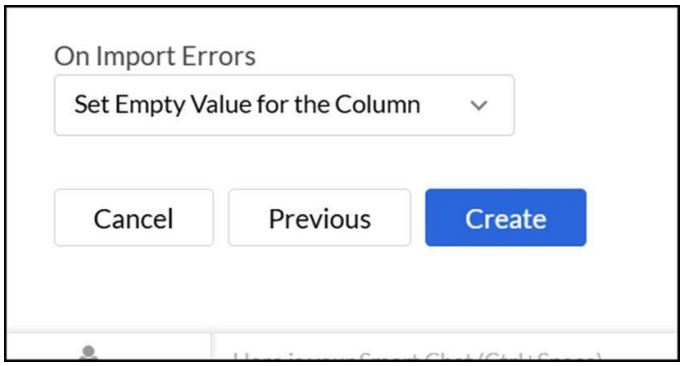


Figure 3.8 – Import error handling set to "Set Empty Value for the Column"; Create button initiates the import

Step Y: Import Success – Data Summary

The Imported Data Summary confirms successful import:

Status: Successfully created BULK-Data-File.HHSCyberSecurityBreaches | Total Columns: 10 | Selected for import: 10 | Total Rows: 1,151 | Successfully Imported: 1,151

All 1,151 records were imported without errors. A tooltip shows "Alaska Department of Health and Social Services" — one of the healthcare entities in the dataset. This complete import enables comprehensive analysis of the breach data.

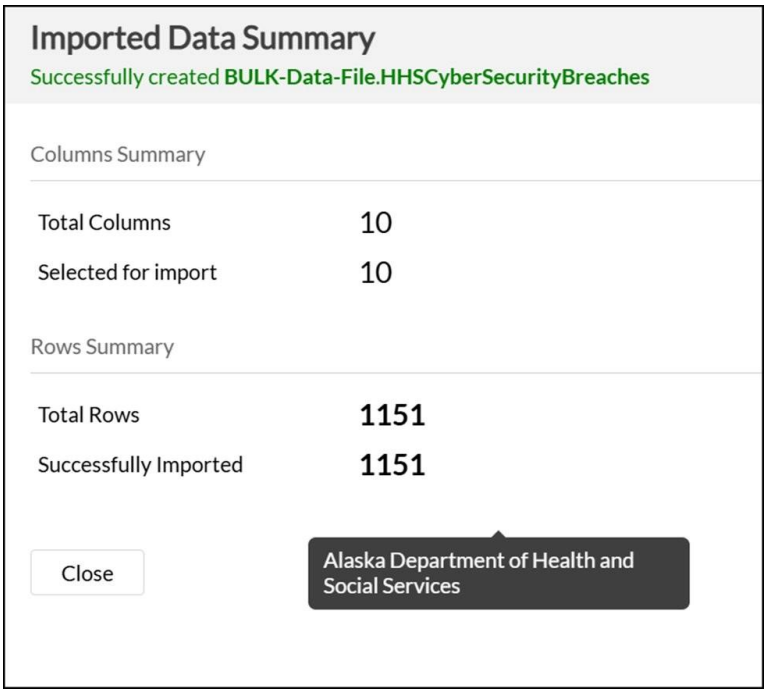


Figure 3.9 – Imported Data Summary confirming all 1,151 rows and 10 columns successfully imported

Step 10: Viewing the Imported Data Table

The HHSCyberSecurityBreaches table view in Zoho Analytics shows all imported records in a spreadsheet-like interface. The first visible row shows Brooke Army Medical Center, TX, Healthcare Provider, 1000 individuals affected, breach date 2009-10-21, Type: Theft, Location: Paper/Films. The toolbar provides options to Import Data, Filter, Sort, Add, Delete, and More. The bottom right confirms "Rows: 1151" and the table has been successfully created.

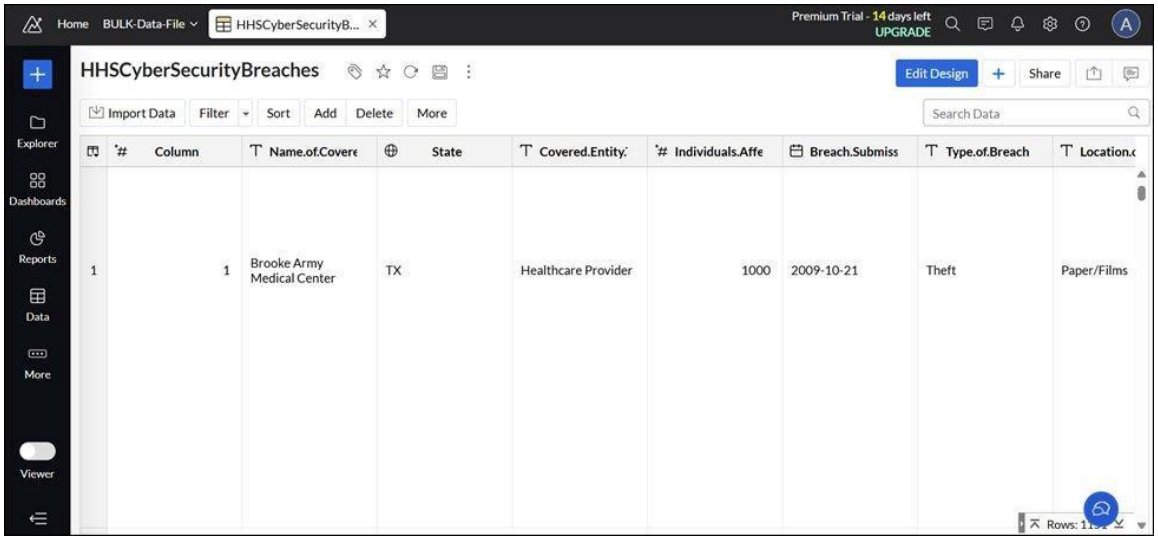


Figure 3.10 – HHSCyberSecurityBreaches data table showing the imported records with all 10 columns visible

Step 11: Creating Reports – Report Type Selection

Clicking the Reports icon in the left sidebar and then the "+" add button reveals the Create New Reports panel with four report types:

Chart View – graphical visualization using various chart types | Pivot View – data summarized in a horizontal/vertical grid | Summary View – data with grouping and summaries | Tabular View – simple tabular format listing

"Chart View" was selected to create visual analytics reports for threat intelligence analysis.

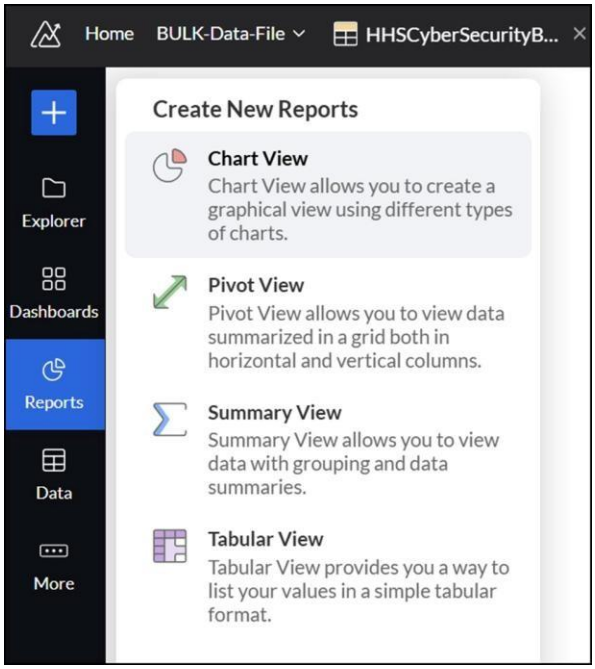


Figure 3.11 – Create New Reports panel showing four report types; Chart View was selected for visual analysis

Step 12: Configuring the Chart – Axis and Dimension Setup

In the Chart View designer, the following configuration was set for a multi-dimensional analysis:

X-Axis: Covered.Entity.Type (Actual) | Y-Axis: Individuals.Affected (Sum) | Color: Type.of.Breach (Actual) | Size: Breach.Submission.Date (Count)

This configuration creates a bubble/scatter-type chart where entity types are on the X-axis, total individuals affected on the Y-axis, breach type determines color, and the count of breach submissions determines bubble size — allowing simultaneous analysis of four dimensions of the breach data.

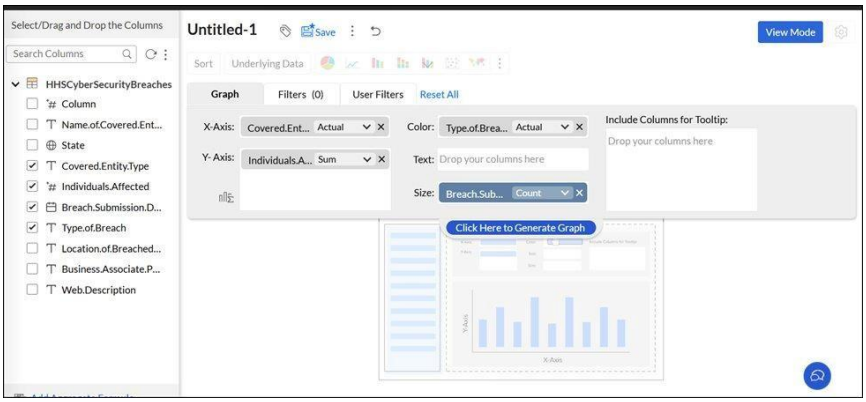


Figure 3.12 – Chart configuration with X-Axis: Covered Entity Type, Y-Axis: Individuals Affected (Sum), Color: Type of Breach, Size: Breach Submission Count

Step 13: Line Chart – Individuals Affected by Entity Type and Breach Type

The generated line chart plots Total Individuals Affected (Y-axis) against Covered Entity Type (X-axis) with different colored lines for each Type of Breach. Key observations from the chart:

Healthcare Providers have the highest overall individuals affected, with several breach types spiking significantly above 8 million affected. Hacking/IT Incident (green line) shows a dramatic spike for Healthcare Providers, reaching over 9 million individuals. Business Associates also show notable breach impacts, particularly for Hacking/IT Incidents. Healthcare Clearing Houses have minimal impact in comparison.

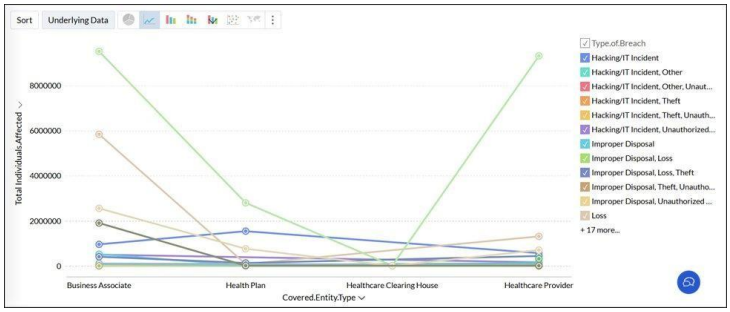
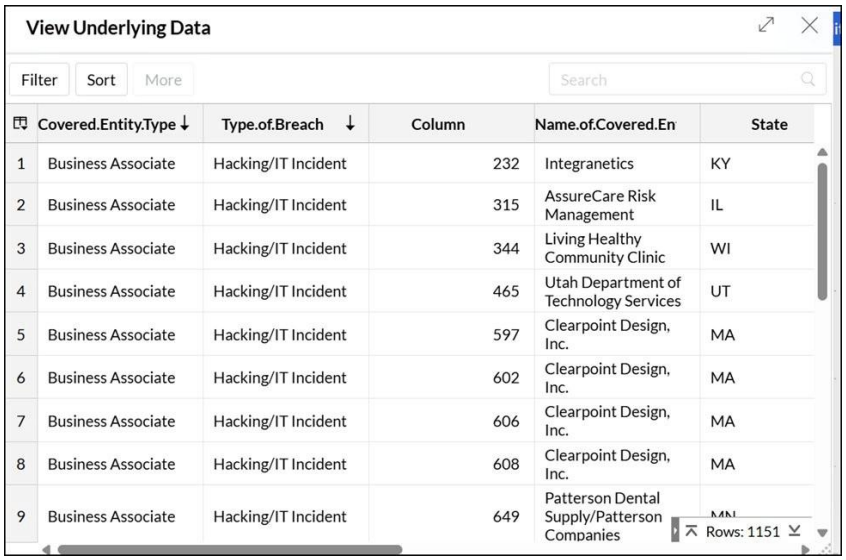


Figure 3.13 – Line chart showing Individuals Affected by Covered Entity Type and Type of Breach; Healthcare Providers show highest impact

Step 14: View Underlying Data

The "Underlying Data" feature was used to drill down into the raw records behind the chart. The table shows detailed records sorted by Covered Entity Type and Type of Breach, starting with Business Associates affected by Hacking/IT Incidents. Notable entries include Integranetics (KY), AssureCare Risk Management (IL), Living Healthy Community Clinic (WI), Utah Department of Technology Services, and multiple Clearpoint Design Inc. (MA) entries. This drill-down capability allows threat analysts to identify specific organizations and incidents behind the aggregate statistics.



	Covered.Entity.Type	Type.of.Breach	Column	Name.of.Covered.En	State
1	Business Associate	Hacking/IT Incident	232	Integranetics	KY
2	Business Associate	Hacking/IT Incident	315	AssureCare Risk Management	IL
3	Business Associate	Hacking/IT Incident	344	Living Healthy Community Clinic	WI
4	Business Associate	Hacking/IT Incident	465	Utah Department of Technology Services	UT
5	Business Associate	Hacking/IT Incident	597	Clearpoint Design, Inc.	MA
6	Business Associate	Hacking/IT Incident	602	Clearpoint Design, Inc.	MA
7	Business Associate	Hacking/IT Incident	606	Clearpoint Design, Inc.	MA
8	Business Associate	Hacking/IT Incident	608	Clearpoint Design, Inc.	MA
9	Business Associate	Hacking/IT Incident	649	Patterson Dental Supply/Patterson Companies	MN

Figure 3.14 – Underlying Data view showing individual breach records for Business Associates with Hacking/IT Incidents

Step 15: Advanced Chart Configuration – Bubble Chart Setup

An advanced bubble chart was configured with additional dimensions:

X-Axis: Covered.Entity.Type (Actual) | Y-Axis: Individuals.Affected (Sum) + Location.of.Breached.Information (Actual) | Color: Column (Sum) | Size: State (Count) | Tooltip: Type.of.Breach (Actual)

This multi-dimensional configuration allows simultaneous visualization of entity type, affected count, breach location type, geographic distribution (state count), and breach type — a powerful intelligence view for identifying high-risk patterns.

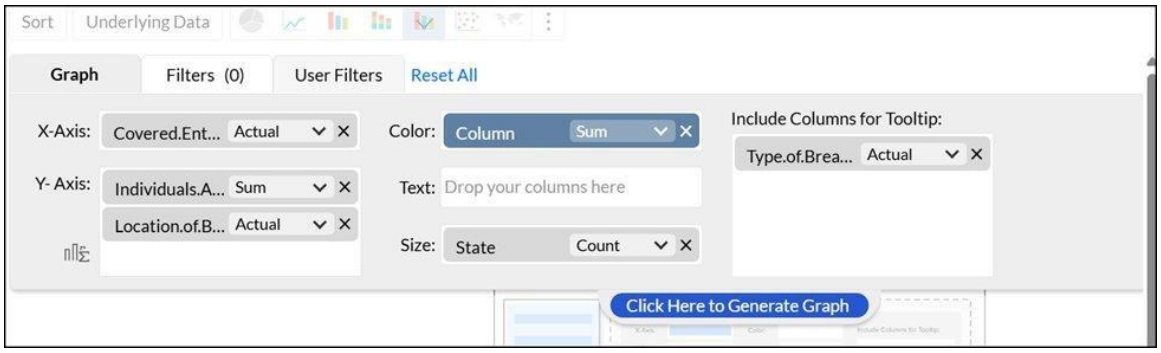


Figure 3.15 – Advanced chart configuration with dual Y-axis, color by column count, size by state count, and breach type tooltip

Step 16: Applying Filters to the Analysis

The Filters tab was used to refine the analysis. A filter on "State (Count)" was applied using Individual Values selection, with options including values 1, 2, 3, 4, 6, 7, 9, 10, 13, 14, 15, 20, 24, 26, 28, 33, 36, 47, 48, 53, 55, 91, 105, and 437. Additional filters were applied on Breach.Submission.Date (Actual) and Type.of.Breach (Actual). These filters allow the analyst to focus on specific subsets of the data, such as states with particular breach counts or specific date ranges.

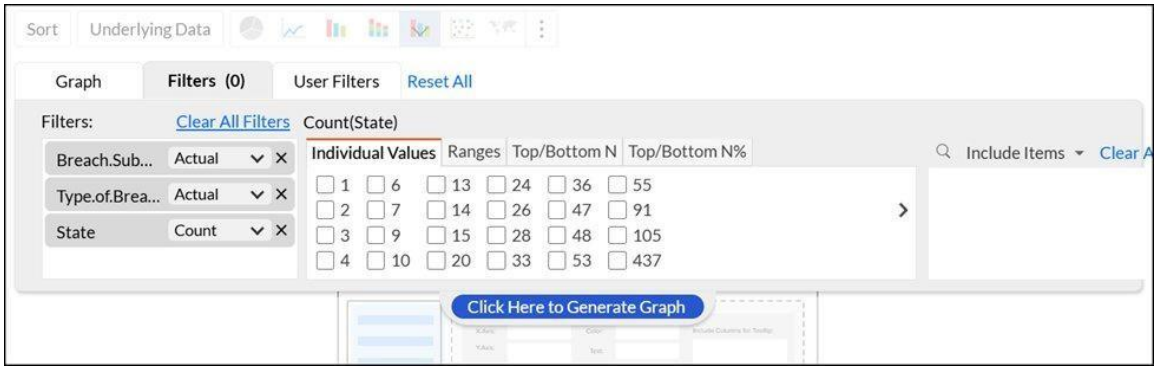


Figure 3.16 – Filters panel showing State Count individual value selection for targeted analysis

Step 17: Final Bubble Chart – Breach Analysis by Entity Type

The final bubble chart visualization displays the multi-dimensional analysis with a tooltip showing detailed information for a Healthcare Provider data point: Covered Entity Type: Healthcare Provider | Total Individuals Affected: 1,855,588 | State Count: 160 | Type of Breach: Theft. The bubble chart clearly shows Healthcare Providers (rightmost cluster) have both the largest bubbles (more states affected) and highest individual impact. The Y-axis shows both Total Individuals Affected and Location of Breached Information categories, enabling comprehensive breach pattern analysis across entity types.

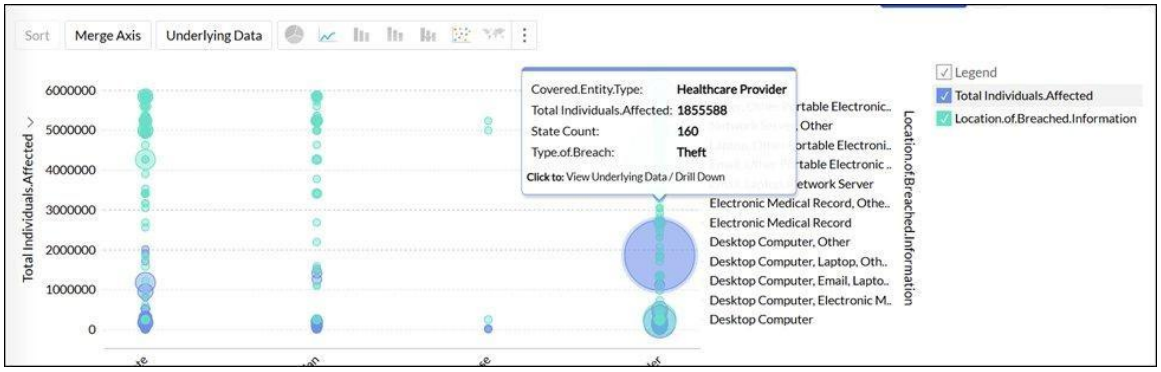


Figure 3.17 – Final bubble chart showing Healthcare Provider tooltip: 1,855,588 individuals affected, 160 states, Type: Theft

Lab Analysis and Conclusions

Lab 1: ACH Analysis Conclusions

The ACH analysis using PARC ACH 2.0.5 successfully demonstrated the structured methodology for evaluating competing hypotheses. By entering two hypotheses (Malicious Patch vs. Misconfigured Permissions) and rating evidence items (targeted payment queries and multi-hospital pattern replication), the weighted inconsistency scores clearly distinguished the more probable explanation:

H:1 (Malicious Patch) scored -1.414 — lower inconsistency, more evidence-consistent. H:2 (Legitimate Patch/Broad Permissions) scored -3.535 — higher inconsistency, more evidence contradicts it. Conclusion: The evidence strongly suggests the EHR vendor was compromised and a malicious patch was deployed. The multi-hospital pattern and targeted financial data queries are highly inconsistent with a simple misconfiguration scenario.

Lab 2: Threat Modeling Conclusions

The Microsoft Threat Modeling Tool 2016 successfully generated a comprehensive threat model for the WebRequestModel architecture (User → Web Application → Web Server → Cloud Storage). Key findings: 44 total threats identified automatically via STRIDE methodology; 4 threats require immediate investigation; 2 threats have existing mitigations. The STRIDE model identified threats including Elevation Using Impersonation, Cross Site Scripting, XML DTD and XSLT Processing threats, and user-defined nonstandard threats. This demonstrates the value of threat modeling as a proactive security measure to identify vulnerabilities before attackers can exploit them.

Lab 3: Data Analysis Conclusions

The Zoho Analytics data analysis of the HHS Cybersecurity Breaches dataset (1,151 records) yielded significant threat intelligence insights:

Healthcare Providers are the most affected entity type, with Hacking/IT Incidents causing the highest number of individuals affected (over 9 million in some cases). Theft is also a significant breach type, with one Healthcare Provider incident alone affecting 1,855,588 individuals across 160 states. Business Associates handling healthcare data also represent significant breach risk, particularly through Hacking/IT Incidents. The bubble chart analysis revealed geographic spread patterns, with some breach types affecting entities across 437 states/territories in the dataset.

These findings demonstrate that organizations handling healthcare data must prioritize protection against hacking and theft — and that business associates represent a significant supply chain risk. The multi-dimensional visualization capabilities of Zoho Analytics enable rapid pattern recognition that would be difficult to achieve through manual data review.